

Image Processing and Pattern Recognition

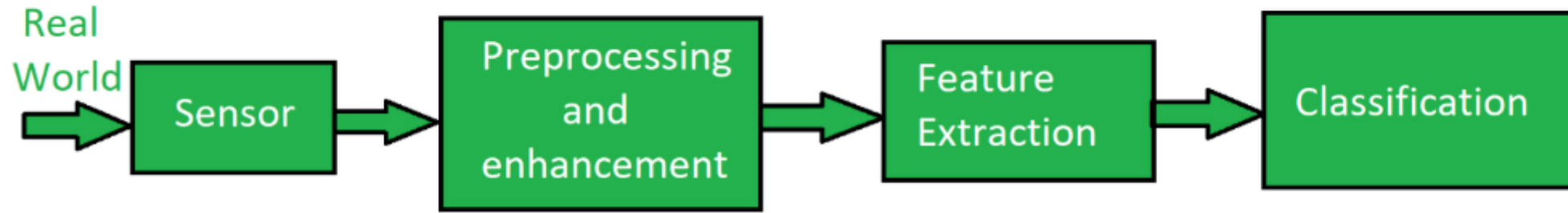
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Pattern Recognition

- Pattern recognition is the automated process of identifying regularities, structures, or trends in data to classify it or make predictions using machine learning algorithms. It acts as a bridge between data analysis and artificial intelligence, enabling computers to interpret images, text, sounds, and complex datasets.

Key aspects of pattern recognition include:

- Methodology:** It utilizes algorithms to analyze input, perform feature extraction, and classify data into categories
 - (e.g., supervised learning) or group similar items together (e.g., unsupervised clustering).
- Applications:** It is widely used in computer vision, facial recognition, speech recognition, bioinformatics, and data mining.
- Process:** The process generally involves four main phases: sensing (data acquisition), pre-processing, feature extraction, and classification.
- Function:** It allows systems to recognize both familiar and unfamiliar objects, even from different angles or in noisy environments.



Pattern Recognition System

Input



Sensing

Segmentation

Feature Extraction

Output



Post Processing

Classification